

Project Name: Talent Tap Pro – Campus Skill-Based Collaboration Engine

1. Current Limitations

While the current prototype successfully demonstrates the core concept of skill-based collaboration, several limitations are acknowledged:

- **Prototype Data Persistence:**
For the purpose of this hackathon demonstration, the system uses browser-based local storage or static data to simulate a database. A production system would require a centralized cloud database (e.g., Firebase or PostgreSQL).
- **Basic Authentication:**
The system uses a simple login/profile creation mechanism. It does not currently integrate with the university's official authentication system (SSO or LDAP).
- **Self-Declared Skills:**
User skills are manually entered and not fully verified. This may lead to inaccurate skill representation and affect matching quality.

2. Technical Limitations

- **Basic Matching Algorithm:**
The current skill-matching system uses simple rule-based logic (keyword matching). It does not yet include advanced AI or recommendation models.
- **No Real-Time Communication:**
The messaging system is either basic or simulated and does not support real-time communication, notifications, or live updates.
- **Scalability Constraints:**
The current architecture is optimized for small-scale demo usage. Large-

scale deployment would require backend optimization, load balancing, and distributed systems.

3. Future Enhancements (Roadmap)

To transform this prototype into a production-ready campus collaboration platform, the following features are proposed

A. Advanced Upskilling & Mentorship System

Development of a structured learning environment:

- **Beginner Inclusion:**
Allow unskilled users to join projects as learners.
- **Outcome:**
Gradual transformation of unskilled users into skilled contributors.

B. Mobile Application Development

Development of a mobile app to improve accessibility:

- Real-time notifications
- Easy project updates
- Better user engagement

C. Enhanced Security & Data Management

Implementation of production-level security:

- Secure authentication (OAuth / SSO)
- Data encryption
- Role-based access control

4. Real-World Deployment Considerations

- **Data Privacy:**

All user data, including profiles, skills, and project information, must be securely stored and encrypted to comply with university data protection policies. Access control mechanisms should ensure that only authorized users can view or modify sensitive data.

- **Adoption Strategy:**

To encourage students to actively use the platform, engagement strategies such as:

- Gamification (badges like “Top Contributor”, “Active Collaborator”)
- Recognition for completed projects
- Incentives for mentorship participation
can be introduced to replace informal platforms like WhatsApp or Discord.

- **Skill Verification Mechanism:**

To ensure trust and reliability, the platform should include methods to validate user skills through:

- Peer endorsements
- Faculty approvals
- Portfolio-based verification

- **System Integration:**

The platform should integrate with existing university systems such as:

- Student databases
- Learning management systems (LMS)
- Authentication systems (SSO)
to ensure seamless user experience.